Certified Robustness for Model-Free PID Tuning by Step-Response Inspection

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Abstract

The SPIN method tunes a PID controller without a model. After each setpoint step, it counts how many turns the control error makes in three phase portraits, and it raises the gains until a count exceeds its limit. The limits were found by calibration, so robustness was not guaranteed. We answer three questions with three theorems. (1) What does a count measure? For any loop, the count of the integral portrait is within half a turn of half the number of setpoint crossings if the lobes of the error shrink; on loops that cancel the plant lags, the lobes shrink whenever $M_s < 26$, and the error never crosses the setpoint if and only if the sensitivity peak satisfies $M_s \leq 1.394$. (2) Can the step show M_s? For any stable digital loop with one degree of freedom, the increments of the error after a unit setpoint step are the impulse response of the sensitivity function, so one finite record gives a lower and an upper bound on M_s without a model. (3) Can SPIN then guarantee robustness? With the upper bound as its acceptance test, SPIN accepts only tunings with $M_s \leq M^*$, and the test never traps it. Simulations on 19 loops confirm the theory.

1 Introduction

Let $L = CP$ be the loop transfer function and $S = 1/(1+L)$ the sensitivity function. Its peak $M_s = \max_\omega |S|$ measures robustness: a value near 1.4 is safe, and a value near 2 indicates poor robustness [2]. Tuning rules reach a chosen M_s by using a model [2, 3]. Once a loop is tuned, M_s is rarely checked, because a check needs either a model or a test that disturbs production [5, 6].

1.1 The SPIN method

SPIN [1] tunes a PI or PID controller with the gains K_i, K_p, K_d without a model. In each iteration, it applies one setpoint step, records the control error e, and forms three phase portraits,

$$\Gamma_0 : (e, \sum e), \quad \Gamma_1 : (\Delta e, e), \quad \Gamma_2 : (\Delta^2 e, \Delta e), \tag{1}$$

where $\sum e$ is the running sum of e and Δ is the difference between consecutive samples. The portraits $\Gamma_0, \Gamma_1, \Gamma_2$ correspond to the frequency bands of K_i, K_p and K_d, which we call bands 0, 1 and 2. The turn count N_k is the number of turns that Γ_k makes around its end point before the error stays within $\pm\delta$ (2% of the step by default). Algorithm 1 states the rule. SPIN needs neither a model nor plant-specific settings [1]. However, its limits $\bar{N}_k$ were found by calibration, and the robustness of the result was not guaranteed. An online simulator lets the reader try SPIN.[1]

Algorithm 1: one iteration of SPIN [1]

```
apply a setpoint step and record the error e
if the record does not settle then
    divide K_i, K_p, K_d by 2, 4, 8                (retreat)
else
    count the turns N_0, N_1, N_2 of Γ_0, Γ_1, Γ_2
    if N_k ≤ N̄_k for every band k then
        raise all gains by one increment           (accept)
    else
        k ← lowest band with N_k > N̄_k
        lower the gain of band k by one increment and
        raise the gains of the bands below k by one
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One increment multiplies or divides a gain by $1/(1-\beta)$. Defaults: $\bar{N} = (0.5, 0.75, 1.0)$, $\delta = 0.02$, $\beta = 0.1$; each gain stays within $[0.001, 10]$ times its initial value (the gain box). The result is the last accepted tuning.

1.2 Contributions

The paper answers three questions, each building on the previous one; each answer is a theorem.

1. *What does a count measure?* (Section 2) A portrait turns by exactly half a turn between two consecutive zeros of its first coordinate, in any linear coordinates (Lemma 1), so the count of Γ_0 is within half a turn of half the number of setpoint crossings (Corollary 1). On loops that cancel the plant lags, the crossings reveal M_s: the error never crosses

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[1] https://robopid-simulator.uceeb.cvut.cz

the setpoint if and only if $M_s \le 1.394$ (Theorem 1). Elsewhere, counts and M_s are not tied.

2. *Can the step show M_s anyway?* (Section 3) Yes. For a digital loop, the record gives the sensitivity function exactly (Lemma 2), and one finite record gives certified lower and upper bounds on M_s (Theorem 2).
3. *Can SPIN then guarantee robustness?* (Section 4) Yes. With the upper bound as its acceptance test, SPIN accepts only tunings with $M_s \le M^*$, the test never traps it (Theorem 3), and the same record predicts which gain to lower (Proposition 2).

Section 5 confirms every statement in simulation.

Related work Frequency responses, sensitivity functions and margins have been estimated from step or relay responses and from closed-loop experiments [7, 8, 9]; PI controllers have been tuned from one setpoint step [14]; and model-free methods optimize performance from repeated experiments or one data set [12, 13, 15]. Here, one ordinary setpoint step certifies M_s inside a model-free tuning rule.

2 Counts Count Crossings

2.1 Every crossing adds half a turn

In each portrait (1), the second coordinate q is the running sum of the first coordinate p; in continuous time, $\dot{q} = p$. The turn count is the net counterclockwise angle swept around the end point, which we place at the origin, divided by 2π. We state the results in continuous time; the sampled counts of SPIN approximate them. For Γ_0, the zeros of $p = e$ are the setpoint crossings; for Γ_1, they are the extrema of e.

Lemma 1 (Half turns) *Let $\dot{q} = p$ with p continuous, let $p(t_1) = p(t_2) = 0$ with $q(t_1)q(t_2) < 0$, and let p not change sign on (t_1,t_2). Then, on $[t_1,t_2]$, the curve (p,q) turns counterclockwise around the origin by exactly half a turn. The same holds for its image under any linear map with a positive determinant, such as the whitening used in [1].*

Proof: On (t_1,t_2), p keeps one sign. Hence the curve stays in one closed half-plane bounded by the q-axis, and q moves monotonically from $q(t_1)$ to $q(t_2)$, that is, from one ray of the q-axis to the opposite ray; the curve avoids the origin, since $q(t) = 0$ only where $p \neq 0$. The angle therefore stays in a closed interval of length π whose ends are the two rays, so it changes by exactly π. If $p > 0$, q increases; if $p < 0$, q decreases; in both cases the turn is counterclockwise. A linear map with a positive determinant maps the half-plane to a half-plane and the two rays to opposite rays, and it preserves the orientation. □

Corollary 1 (Counts and crossings) *Let $p(0) \neq 0$, and let p change sign at $t_1 < \cdots < t_m$, $m \ge 0$, before the error stays within $\pm\delta$. Let $q(t_j)$ alternate in sign, and let $q(t_1)$ have the sign of p before t_1. Then the turn count of (p,q), and of its image under any linear map with a positive determinant, lies in $(\frac{m-1}{2}, \frac{m+1}{2})$. For Γ_0, these conditions hold if the areas of the lobes of e between crossings decrease strictly.*

Proof: For $m = 0$, the curve stays in one open half-plane, so its angle changes by less than π. For $m \ge 1$, by Lemma 1, each of the $m-1$ intervals between sign changes adds π. Before t_1 and after t_m, the curve stays in one open half-plane; by the sign conditions, it ends (before t_1) or starts (after t_m) on the ray from which it turns counterclockwise, so each end adds an angle in $(0,\pi)$. For Γ_0, $q(t_j) = -\int_{t_j}^{\infty} e\,dt$ is an alternating sum of lobe areas; if the areas decrease, the sum has the sign of its first term. □

2.2 When crossings reveal M_s

Lambda, IMC and SIMC tunings cancel the plant lags [3], e.g., a PI controller $K_c(1 + 1/(T_i s))$ with $T_i = T$ on the plant $Ke^{-sL_d}/(Ts+1)$, or a series PID controller with $T_i = T_1$ and $T_d = T_2$ on a plant with two lags. The loop is then

$$L(s) = \frac{k\,e^{-sL_d}}{sL_d}, \qquad k = \frac{KK_cL_d}{T_i}. \tag{2}$$

Proposition 1 (Exact robustness) *The loop* (2) *is stable for $0 < k < \pi/2$. Its gain margin is $\pi/(2k)$, and its poles are $W_j(-k)/L_d$, where W_j are the branches of the Lambert W function [10]. Its peak M_s is a strictly increasing function of k, given by*

$$k = x\sin x - x^2\cos x, \qquad \frac{1}{M_s^2} = 1 - \frac{2k\sin x}{x} + \frac{k^2}{x^2}, \tag{3}$$

where $x = \omega_p L_d \in (0,\pi/2)$ is the normalized peak frequency.

Proof: The phase of L is $-\pi/2 - x$. It first reaches $-\pi$ at $x = \pi/2$, where $|L| = 2k/\pi < 1$; since $|L| = k/x$ decreases, $|L| < 1$ at every later crossing of $-\pi$, so the loop is stable (Nyquist). The poles solve $sL_de^{sL_d} = -k$. Let $h(x) = |1+L|^2 = 1 - 2k\sin x/x + k^2/x^2$. Then $h'(x) = 2k(\varphi(x) - k)/x^3$, where $\varphi(x) = x\sin x - x^2\cos x$ increases on $(0,\pi)$ and $\varphi(\pi/2) = \pi/2 > k$. Hence, on $(0,\pi)$, h has a single minimum $x^* \in (0,\pi/2)$. On $[\pi,2\pi]$, $h \ge 1$; for $x \ge 2\pi$, $h \ge (1-k/x)^2 > (1-2k/\pi)^2 = h(\pi/2)$. So x^* is the global minimum, which gives (3). By the envelope theorem, $\frac{d}{dk}h(x^*) = \frac{2}{x^{*2}}(k - x^*\sin x^*) = -2\cos x^* < 0$. □

SIMC with $\tau_c = L_d$ and $T \le 8L_d$ cancels the lag and gives $k = \frac{1}{2}$. Hence the gain margin is π and, by (3), $M_s = 1.59$, which are exactly the values in [4, Table 2].

After a unit setpoint step, $e = 1$ on $[0,L_d]$ and $\dot{e}(t) = -(k/L_d)\,e(t-L_d)$. Let $c_1 < c_2 < \ldots$ be the setpoint crossings, let $c_0 = 0$, let lobe i be the interval $[c_i, c_{i+1}]$, and let $E_0 = 1$.

Theorem 1 (Counts on lag-cancelled loops) *For the loop (2) with $0 < k < \pi/2$, the following hold.*

(a) The error never crosses the setpoint if and only if $k \le 1/\mathrm{e}$, that is, if and only if $M_s \le M_s(1/\mathrm{e}) = 1.394$. For $k > 1/\mathrm{e}$, it crosses infinitely often.

(b) Lobe i contains exactly one extremum E_i, at $c_i + L_d$.

(c) $|E_{i+1}| \le r(k)\,|E_i|$, where $r(k) = k^2/2$ for $k \le 1$ and $r(k) = k - \frac{1}{2}$ for $k \ge 1$.

(d) For $k < 3/2$, the lobe areas decrease strictly, so Corollary 1 holds for Γ_0: N_0 is within half a turn of half the number of crossings.

Proof: (a) The record is the fundamental solution of the delay equation. It is positive if and only if $s + (k/L_d)e^{-sL_d} = 0$ has a real root, that is, if and only if $k \le 1/\mathrm{e}$; for $k > 1/\mathrm{e}$, all solutions oscillate [11]. The value of M_s follows from Proposition 1. (b) The sign of $\dot e(t)$ is that of $-e(t-L_d)$. By induction over the lobes, e moves away from zero on $(c_i, c_i + L_d)$, where $e(t-L_d)$ lies in lobe $i-1$; hence lobe i is longer than L_d. On $(c_i + L_d, c_{i+1} + L_d)$, e moves toward zero. For $i = 0$, the extremum is the plateau $e = 1$ on $[0, L_d]$. (c) On lobe i, $|e| \le |E_i|$. After the extremum, $|\dot e| \le (k/L_d)|E_i|$, so e needs at least L_d/k to reach zero, and $|e(t)| \le |E_i| \min\{1, k(c_{i+1} - t)/L_d\}$ on the last L_d of the lobe (for $t \ge c_i + L_d$ by integrating the bound on $|\dot e|$; for earlier t, $k(c_{i+1} - t)/L_d > 1$). Since $e(c_{i+1}) = 0$, $E_{i+1} = -\frac{k}{L_d}\int_{c_{i+1}-L_d}^{c_{i+1}} e\,dt$, and integrating the bound gives $r(k)|E_i|$. (d) Integrating $\dot e$ from $c_i + L_d$ to $c_{i+1} + L_d$ gives the area $\frac{L_d}{k}(|E_i| + |E_{i+1}|)$ of lobe i, because the extrema alternate in sign. By (c), $|E_{i+2}| < |E_i|$ if $r(k) < 1$, that is, if $k < 3/2$, where $M_s = 26.2$. □

Thus N_0 reveals the number of crossings, and the crossings reveal M_s (Fig. 1).

2.3 When they do not

Outside lag-cancelled loops, crossings and M_s are not tied. Consider a first-order plant with $T = 0.3L_d$, controlled by PI with $T_i = 8T$ and a gain margin of 1.624. Its M_s is 2.62, yet its error never crosses the setpoint by more than 2%, because the oscillation rides on a slow decay (2.61 for the digital loop of Fig. 2). Conversely, overshoot caused by a controller zero can produce crossings at a low M_s. Counts can guide tuning but cannot certify it; the same step, however, contains M_s.

3 One Step Reveals M_s

A digital controller runs with the sampling period T_s. Seen at the samples, the loop it closes is linear and time invariant. From here on, $L(z)$, $P(z)$, $S = 1/(1+L)$ and $M_s = \max_{\theta\in[0,\pi]} |S(e^{j\theta})|$, with $\theta = \omega T_s$, refer to this digital loop, the loop that actually runs.

Lemma 2 (Exact identity) *Let the loop be stable, have one degree of freedom, and have a strictly proper L (the measurement precedes the control action). Let e_n be the error after a unit setpoint step at $n = 0$, applied from rest and without disturbances, so that $e_{-1} = 0$ and $e_0 = 1$, and let $a_n = \Delta e_n = e_n - e_{n-1}$. Then*

$$S(e^{j\theta}) = \sum_{n\ge 0} a_n\, e^{-jn\theta}. \tag{4}$$

Proof: The z-transform of e is $S(z)/(1 - z^{-1})$, so $S = (1 - z^{-1})\mathscr{Z}\{e\}$. In a stable loop, a_n decays exponentially, so the series converges on the unit circle. □

A real record is finite; it cannot show the remaining variation of the error after its end.

Assumption 1 (Long enough record) *The loop is stable, and the record $e_0, \dots, e_N$ satisfies $\sum_{n>N} |a_n| \le \tau$ for a known τ.*

In a stable loop, $|a_n| \le c\rho^n$ with $\rho < 1$, so τ falls geometrically with the record length.

Theorem 2 (Certified bounds from one record) *Let the conditions of Lemma 2 and Assumption 1 hold, and let $\hat S(\theta) = \sum_{n=0}^{N} a_n e^{-jn\theta}$. Let G be a uniform grid on $[0, \bar\theta]$, $\bar\theta \le \pi$, that has spacing h and contains both ends. For any n_c, let*

$$J = \textstyle\sum_{n=0}^{N} |n - n_c|\,|a_n|, \quad B = \sum_{n=1}^{N-1} |a_{n+1} - a_n| + |a_N| + 2\tau.$$

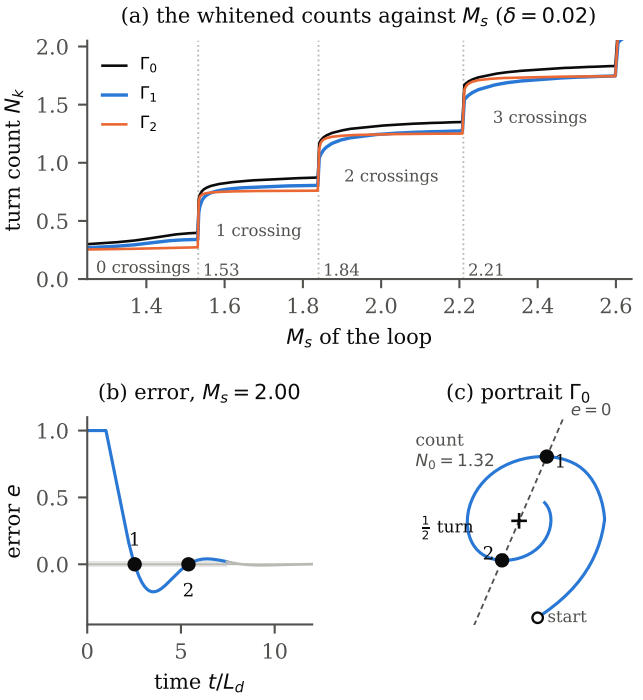


Figure 1: The lag-cancelled loop. (a) Whitened counts versus M_s ($\delta = 0.02$); N_0 stays within half a turn of half the number of crossings (Corollary 1). (b) A record with $M_s = 2.00$ and two crossings. (c) Its portrait Γ_0 turns by exactly half a turn between the crossings (Lemma 1); the radius is compressed but the angle is not.

Algorithm 2: certified SPIN; line 7 of Algorithm 1 is replaced by

7a: compute $\overline{M}$ by Theorem 2, with $\bar{\theta}$
from $1+B/\sin(\bar{\theta}/2) = M^*$ ($\bar{\theta}=\pi$ if no solution)
and $h = \bar{\theta}/\lceil \bar{\theta}J/(2\eta)\rceil$, so that $hJ/2 \le \eta$
7b: **if** $\overline{M} \le M^*$ **then** raise all gains by one increment (accept)
7c: **else** predict by Proposition 2 the M_s after
each band move (lower one band, raise those below)
7d: make the band move of the highest band predicted
to lower M_s, at most $n_{\max}$ times between
acceptances; otherwise lower all gains by one increment

Defaults: $M^* = 1.7$, $\tau = 10^{-3}$, $\eta = 2\cdot 10^{-3}$, $n_{\max} = 10$.

Then $\underline{M} \le M_s \le \overline{M}$, where

$$\underline{M} = \max_{\theta\in G} |\hat{S}(\theta)| - \tau,$$
$$\overline{M} = \max\Big\{ \max_{\theta\in G} |\hat{S}(\theta)| + \tfrac{hJ}{2} + \tau,\ 1 + \tfrac{B}{\sin(\bar{\theta}/2)} \Big\}, \quad (5)$$

and the second term is omitted if $\bar{\theta} = \pi$. If τ equals the true remaining variation, $N\to\infty$, $h\to 0$ and $\bar{\theta}=\pi$, then both bounds converge to M_s.

Proof: By (4), $|S-\hat{S}| \le \sum_{n>N} |a_n| \le \tau$ at every θ, which gives $\underline{M}$. On $[0,\bar{\theta}]$, $|\hat{S}|$ equals $|e^{jn_c\theta}\hat{S}|$, whose derivative is bounded by J. Every θ lies within $h/2$ of a grid point, which gives the first term of $\overline{M}$. On $[\bar{\theta},\pi]$, let $z = e^{-j\theta}$ and $Z_n = \sum_{m=1}^{n} z^m$, so that $|Z_n| \le 2/|1-z| = 1/\sin(\theta/2)$. Summation by parts gives $\sum_{n\ge1} a_n z^n = \sum_{n\ge1}(a_n - a_{n+1})Z_n$. Since $\sum_{n\ge N}|a_n - a_{n+1}| \le |a_N| + 2\tau$, we obtain $|S-a_0| \le B/\sin(\theta/2)$ with $a_0 = 1$. Convergence follows because, for fixed n_c, J stays bounded as $N\to\infty$, so $hJ\to 0$, and the remaining variation tends to zero. □

Every quantity in (5) except τ is computed from the record, and all grid values follow from one chirp z-transform. The two terms of $\overline{M}$ split the frequency axis: (1) on $[0,\bar{\theta}]$, the grid term bounds $|S|$ by its grid values; (2) on $[\bar{\theta},\pi]$, the second term needs no grid: for a smooth record, B is small and bounds all high frequencies at once.

4 Never Accept a Fragile Tuning

Algorithm 2 changes only line 7 of SPIN. The counts still shape the tuning, and the bound decides whether a tuning is accepted. After a rejection, it uses two kinds of moves: (1) the band move of line 10 of Algorithm 1, and (2) the uniform back-off, the reverse of line 7.

Theorem 3 (Certified SPIN) *Let the plant be stable, let $K_iP(1) > 0$, and let every record tested in line 7a meet the conditions of Lemma 2 and Assumption 1. For (iii) and (iv), also let no gain be clipped by the gain box, and let $\beta < \frac{1}{2}$.*

(i) Every tuning that passes the test of line 7b, and hence the result of Algorithm 2, has $M_s \le M^*$.

(ii) For a controller C, let $\mu = \min\{0, \inf_{0<\theta\le\pi} \mathrm{Re}\, L(e^{j\theta})\}$. For $0 < \lambda < 1/|\mu|$, the loop with the controller λC is stable, and

$$M_s(\lambda C) \le \frac{1}{1-\lambda|\mu|}. \quad (6)$$

(iii) If $M^* > 1+2\tau+\eta$, every run of consecutive rejections by the test of line 7b ends, either in an acceptance or in a move of Algorithm 1 (line 3 or line 10). The run contains at most $n_{\max}$ band moves, and each stretch of uniform back-offs in it has at most $\lceil \ln \frac{|\mu|M'}{M'-1} / \ln \frac{1}{1-\beta} \rceil$ of them, where $M' = M^* - 2\tau - \eta$ and μ belongs to the controller at the start of the stretch.

(iv) Every move except an acceptance lowers $\Phi = \ln K_i + 2\ln K_p + 4\ln K_d$ (without the last term for PI) by at least $\ln \frac{1}{1-\beta}$. Hence the gains never return to earlier values without an acceptance in between.

Proof: (i) This follows from Theorem 2. (ii) L is bounded on the unit circle except near $z=1$. There, the integral term has a bounded real part, because P is analytic at $z=1$ and $\mathrm{Re}\, 1/(e^{j\theta}-1) = -\frac{1}{2}$; so μ is finite. Then $|1+\lambda L| \ge 1 + \lambda\,\mathrm{Re}\, L \ge 1 - \lambda|\mu| > 0$, which gives (6). The Nyquist curve of λL lies in $\mathrm{Re} \ge -\lambda|\mu| > -1$, also on a small enough indentation outside $z=1$, where $K_iP(1) > 0$ keeps $\mathrm{Re}\, L > \mu$. It therefore cannot encircle -1, and since L has no poles outside the unit disc, the loop is stable. (iii) Line 7d allows at most $n_{\max}$ band moves. In a stretch of uniform back-offs from a controller C, $\lambda = (1-\beta)^j$ after j back-offs, and (6) gives $M_s \le M'$ for the stated j. In (5), the grid term is then at most $M_s + \tau + \eta + \tau \le M^*$, and the second term is at most M^* by the choice of $\bar{\theta}$; so the test passes unless the counts reject first. (iv) With $w = (1,2,4)$, a band move changes Φ by $\ln\frac{1}{1-\beta}(\sum_{j<k} w_j - w_k) \le -\ln\frac{1}{1-\beta}$. A uniform back-off changes it by $-7\ln\frac{1}{1-\beta}$, and line 3 by $-17\ln 2 < -\ln\frac{1}{1-\beta}$ for $\beta < \frac{1}{2}$. Returning to the same gains requires $\sum\Delta\Phi = 0$, hence an acceptance, which raises Φ. □

So the test never traps SPIN: it certifies a tuning or returns control to the original rules. Part (i) holds with or without the gain box. Part (iv) allows a drift in which Φ decreases forever; this is a property of the counts, stopped by the gain box (Section 5).

A uniform back-off is safe by (ii) but worsens load rejection; a band move is often cheaper, and the record predicts what each move does.

Proposition 2 (Prediction from one step) *For any controller C' on the same plant, $S' = S/(S+(1-S)C'/C)$ exactly. If the record gives $\hat{S} = S+\varepsilon$, the predicted $\hat{S}'$ has the error $\hat{S}' - S' = \varepsilon\,(C'/C)/(D\hat{D})$, where $D = S+(1-S)C'/C$ and $\hat{D}$ is D with $\hat{S}$, wherever $D,\hat{D} \ne 0$. The formula predicts the frequency response of the new loop, not its stability.*

Proof: $L' = LC'/C$ with $L = (1-S)/S$; subtracting the two expressions gives the error. □

The prediction only selects the move; the next iteration tests it, so safety never depends on the prediction.

5 The Theory at Work

Setup We compare SPIN as published with certified SPIN (Algorithm 2). Both run 200 iterations with a derivative filter ratio of 10 and the gain box of [1]. Because the box ties the result to the initial tuning, every loop runs from two initial tunings: SIMC on a half-rule model with τ_c equal to three times its dead time (cautious start), and the same tuning with all gains doubled (bold start). The 19 test loops are:

1. The four plants of [1], with PI and with PID,

$$P_1 = \frac{e^{-s}}{(10s+1)(s+1)^3}, \qquad P_2 = \frac{1.25\,e^{-8s}}{(5s+1)^4},$$
$$P_3 = \frac{e^{-10s}}{(2s+1)(s+1)^3}, \qquad P_4 = \frac{e^{-4s}}{(8s+1)^6}.$$

2. Four plants that place the oscillation in a chosen band: the delay-dominant $e^{-5s}/(s+1)$ and the lag-dominant $e^{-s}/(10s+1)$ and $e^{-0.5s}/((10s+1)(s+1))$ with PI; the fast $e^{-0.1s}/((s+1)(0.5s+1))$ with PID.
3. The family $e^{-L_d s}/((s+1)(0.5s+1)(0.25s+1))$ with PID and $L_d \in \{0.25, 0.5, 1, \ldots, 16\}$, whose dead-time share runs from 0.125 to 0.90.

P, I and D act on the error. Let T_a be the dead time plus the lag time constants of the plant and T_f the derivative filter time constant; the controller runs at $T_s = \min(T_a/100, T_f/20)$ for PID and $T_a/1000$ for PI, with exactly discretized lags and the delay rounded to whole samples. Each record covers $15\,T_a$. All values of M_s are those of this digital loop, computed from its exact L. Because SPIN ends in a cycle between a few tunings, we report the most aggressive tuning accepted in the last 60 iterations.

Counts (Section 2) On the lag-cancelled loop with $1/\mathrm{e} < k < 3/2$, for all 126 values of k tested, (1) bound (c) of Theorem 1 held for every extremum, with equality for the first one when $k \geq 1$; (2) the lobe areas decreased, and Γ_0 turned by half a turn between crossings to within 10^{-6}, also after a linear map; (3) the whitened count of SPIN satisfied Corollary 1; (4) (3) matched a direct search for M_s to six digits.

Bounds (Section 3) For all 146 tunings accepted in the runs below, a record four times longer showed the true remaining variation $\sum_{n>N}|a_n| \leq 2.8 \cdot 10^{-5}$, so Assumption 1 held with $\tau = 10^{-3}$. The identity (4) held up to this variation, and every true M_s lay within its bounds, with $\overline{M} - M_s \leq \eta + \tau + 2.8 \cdot 10^{-5}$. On short records the bounds are wide, yet they contain M_s and tighten as the record grows (Fig. 2(b)).

Certified SPIN (Section 4; Table 1, Fig. 3)

- *Safety.* Published SPIN ended above $M^* = 1.7$ on 9 of the 19 loops from the cautious start (up to 2.01) and on 8 from the bold start (up to 2.09); which loops failed depended on the start, because the gain box often set the result. Certified SPIN ended at or below M^* on all loops from both starts (at most 1.69), as Theorem 3(i) guarantees.

Table 1: M_s of the digital loop for the most aggressive tuning that published and certified SPIN accept, from the cautious and the bold start, and the change of the load-step IAE (cautious start; –: tuning unchanged).

		published		certified		
		caut.	bold	caut.	bold	ΔIAE
P_1	PI	1.46	1.46	1.46	1.46	–
	PID	1.66	**1.84**	1.66	1.61	–
P_2	PI	1.64	1.67	1.64	1.67	–
	PID	1.59	**1.78**	1.59	1.66	–
P_3	PI	1.67	1.63	1.67	1.63	–
	PID	**1.81**	**1.74**	1.69	1.60	+23%
P_4	PI	1.55	1.60	1.55	1.60	–
	PID	1.61	1.55	1.61	1.55	–
delay-dom.	PI	1.68	1.63	1.68	1.63	–
lag-dom.	PI	**2.01**	1.69	1.67	1.69	+52%
2 lags	PI	1.44	1.47	1.44	1.47	–
fast	PID	**1.89**	**1.98**	1.66	1.69	0%

- *Without the gain box.* Published SPIN exceeded M^* on 12 of the 18 loops that accepted a tuning in their last 60 iterations (up to 2.05), whereas certified SPIN stayed at or below M^* on all 18 (up to 1.68).
- *Back-off and prediction.* Along the uniform back-off from each most aggressive published tuning, the loop stayed stable and (6) held; the predicted M_s after every band move matched the true value to within 10^{-7}.
- *No trapping.* Runs of rejections by the test contained at most 4 band moves, fewer than $n_{\max}$, and every loop had at least 7 acceptances in its last 60 iterations. The one exception is $L_d = 16$ without the box, where under either rule the counts reject in a repeating pattern that lowers K_d every round: the drift that Theorem 3(iv) allows.
- *Cost.* The integrated absolute error (IAE) after a unit load step at the plant input rose by 7.8% on average (6.4% from the bold start).

6 Conclusion

The three questions of the introduction have clear answers.

1. A count measures setpoint crossings; on lag-cancelled loops, an error that never crosses the setpoint means $M_s \leq 1.394$ (Theorem 1).
2. The setpoint step that SPIN already records gives the sensitivity function of a digital loop exactly, and one finite record certifies M_s without a model (Theorem 2); so does every logged setpoint change made from rest.
3. With this certificate as its acceptance test, SPIN accepts only tunings with $M_s \leq M^*$, and the test never traps it (Theorem 3).

The theorems assume a linear loop and a long enough record; noise and real loops are left to a follow-up. Scripts: `https://github.com/ottapav/robopid-simulator`

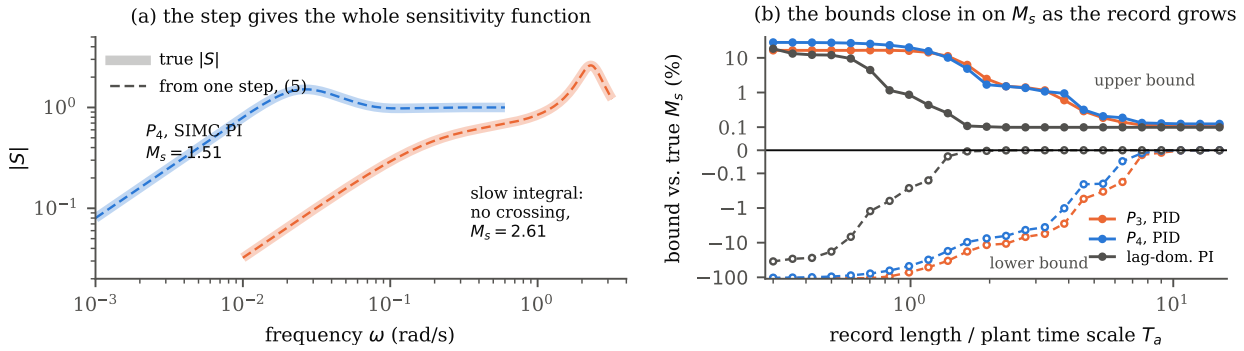


Figure 2: (a) $|S|$ from one setpoint step (dashed) lies on the true $|S|$ (thick), also for the slow-integral loop whose peak the counts miss (Lemma 2). (b) The bounds (5) relative to the true M_s versus the record length, with τ the true remaining variation (Theorem 2).

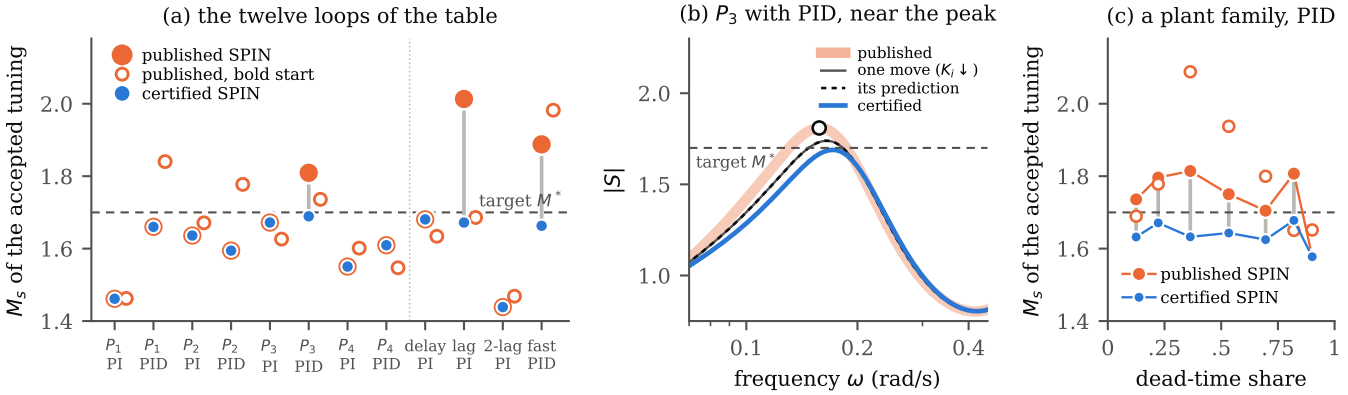


Figure 3: Published versus certified SPIN ($M^* = 1.7$). (a) The loops of Table 1. (b) P_3 with PID: published tuning ($\overline{M} = 1.81$, ○); one band move lowering K_i ($M_s = 1.739$, as predicted by Proposition 2); certified tuning (1.689). (c) The plant family versus its dead-time share.

(`docs/ECC27_certified_SPIN`).

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